A Terminal Is All You Need: ServiceNow Shows Simple Agents Beat Complex Ones for Enterprise Tasks

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ServiceNow researchers built StarShell, a minimal coding agent with nothing but a terminal and a filesystem, then tested it against GUI-based web agents and tool-augmented agents using Model Context Protocol (MCP, a standard for exposing curated API tools to models).

They ran all three agent types across 729 tasks on ServiceNow, GitLab, and ERPNext with four frontier LLMs. Terminal agents matched or beat web agents in 7 of 12 platform-model combinations. MCP agents scored worst, hitting just 11.5-16.7% on ServiceNow despite 93 available tools, because rigid tool schemas couldn't express arbitrary API operations. The best config, Gemini 3.1 Pro with a terminal, reached 77.5% overall at $0.09 per task.

Web agents cost 4-9x more because every interaction requires processing large accessibility trees and screenshots. On ERPNext with Opus 4.6, web agents cost $6.49 per task versus $0.72 for terminal, with only a modest accuracy edge (81.6% vs. 76.8%). When terminal agents got a persistent "skills" directory to store notes from previous tasks, costs dropped another 43.7% on ServiceNow while accuracy rose 3.6%. The agent learned to skip re-exploring APIs it had already figured out.

This reinforces what earlier multi-agent research found: complexity for its own sake rarely pays off. The industry is investing heavily in MCP registries, browser automation, and multi-agent orchestration, but a strong model with direct API access can discover endpoints, write code, and iterate through a simple terminal. ServiceNow, whose entire business is enterprise workflow automation, publishing this result carries weight.

Sources:

- Terminal Agents Suffice for Enterprise Automation (arXiv)
- HuggingFace Daily Papers (#1 Paper, April 2)
- Model Context Protocol (Anthropic)
- Beyond Browsing: API-Based Web Agents (ACL 2025)

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